WHEN DOES MULTI-TEACHER ON-POLICY DISTILLATION MAKE JOINT PROGRESS?

Anonymous authors
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ABSTRACT

Multi-teacher on-policy distillation (MOPD) aims to combine capabilities from several teachers in one student. Successful transfer from each teacher separately does not establish that their combined updates improve every task. We derive how teacher log-ratio signals on retained tokens produce the student's task gradients, then separate the mean update's direction and rate from uncertainty due to finite sampling. An exact quadratic example has no task conflict or sampling noise but unequal convergence rates. For losses on fixed student-generated states and a frozen optimizer scaling, we bound the probability of any non-descent first-order direction when the mean update improves every task to first order. The analysis distinguishes harmful mean interactions, which allocation at fixed weights cannot correct in this local model, from sampling errors in otherwise useful directions. Gradient-Preconditioned Adaptive Sampling (GPAS) minimizes an estimated total-variance criterion; an analytical example shows why this need not minimize uncertainty in task progress. We specify a four-domain study with separate-distillation references, controlled updates from common checkpoints, and repeated training comparisons. The evaluation distinguishes fixed-state loss, fresh-policy loss, downstream capability, and computation. **[TODO: Add the observed integration gap, diagnostic prediction accuracy, and measured capability and compute effects after completing the study.]**

1 INTRODUCTION

Multi-teacher on-policy distillation (MOPD) provides a practical way to combine capabilities in one language model. The student generates responses, and each response is scored by the teacher assigned to its task (Ma et al., 2026). This allows specialist teachers to be developed separately and their feedback to be combined during student training. The same approach can add a new capability while using an earlier model as a teacher to preserve existing ones (Liu et al., 2026). The intended outcome is a student that acquires useful capabilities across tasks. A decrease in average teacher loss alone does not establish that outcome.

Three questions must be distinguished. Can this student acquire each capability through separate distillation? Can one shared model attain those capability levels together? Can training reach them with the available computation? Open-MOPD uses separately distilled students as references and identifies capability imbalance in joint training (Gao et al., 2026). These references establish what the student learns separately; a gap in joint training can still reflect task exposure, weak usable teacher signals, interacting updates, or other optimization effects.

We study a local question about the training process:

> Given capabilities that a student can acquire separately, when does a shared MOPD update improve task objectives together, what limits its rate, and when can more precise estimates help?

Figure 1 separates three cases. Exact, nonconflicting updates can still make slow progress. A combined mean gradient can oppose one task's loss gradient. Or a useful mean direction can be estimated too imprecisely. Additional samples can improve precision; under fixed task weights and optimizer scaling, they cannot change the mean direction or its deterministic local rate.

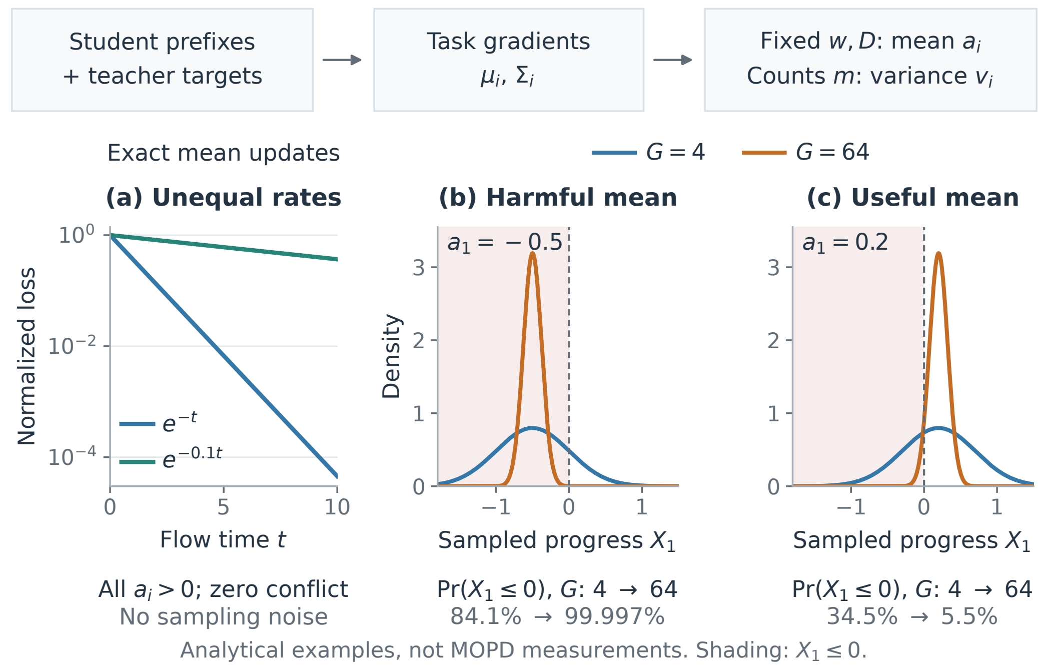

Figure 1: Distinct limits on joint progress, illustrated analytically. (a) Orthogonal, noiseless task gradients yield unequal loss decay rates (Equation 11). (b,c) Sampling concentrates first-order progress X_1 around a harmful or useful mean. With $D = I$, $w_1 = w_2 = 1/2$, independent Gaussian microbatch gradients, $\Sigma_1 = \Sigma_2 = I$, and $m_1 = m_2 = G/2$, $X_1 \sim \mathcal{N}(a_1, 1/G)$. Set $\mu_1 = (1,0)^\top$ and $\mu_2 = (-2,1)^\top$ in (b), or $(-0.6,1)^\top$ in (c). Shading marks $X_1 \le 0$. The sample-size comparison changes total budget G; GPAS instead reallocates a fixed budget and need not reduce every task's directional uncertainty. These are analytical examples, not measured MOPD results.

Existing work motivates testing these possible causes. CaMOPD studies counteracting recovery and preservation gradients (Chen et al., 2026); D^3-MOPD allocates samples using the remaining teacher gap and descent velocity (Sun et al., 2026). Slow alignment has also been observed when the training states are held fixed (Fu et al., 2026). These findings do not determine whether sampling uncertainty limits a particular MOPD run. Dense top-k supervision already computes multiple token contributions at each prefix, while prompts and student trajectories continue to produce variable gradients (Ma et al., 2026).

We derive how teacher log-ratios on retained tokens produce task gradients, then analyze the mean progress and directional uncertainty they imply. A zero-conflict example separates common descent from fast learning; a classical one-sided probability bound relates sampling to first-order reliability. The analysis fixes the student-generated states and optimizer scaling. We measure state-distribution changes and actual AdamW updates separately. Local progress is a diagnostic; successful training can temporarily increase a task loss.

Gradient-Preconditioned Adaptive Sampling (GPAS) provides an allocation intervention for this analysis. It applies Neyman allocation to gradient variance after optimizer scaling and preserves the specified task mean weights. Its per-step rule minimizes an estimate of total update variance, a conservative measure of uncertainty in task progress. We give an analytical example in which the variance-minimizing allocation has greater directional uncertainty than another allocation. A four-domain protocol tests the diagnostics against controlled updates, separately acquired capabilities, and end-to-end training efficiency. **[TODO: Replace the protocol contribution with measured findings after the diagnostic and training comparisons are complete.]**

2 Capability Integration and Local Task Losses

2.1 Separately acquired capabilities as references

Let $C_i(\theta)$ be the student's capability score on task i, with larger values indicating better performance. Separately distilling each teacher into the same student architecture gives reference scores r_i. A capability-integration target is a shared student satisfying

$$C_i(\theta) \geq r_i - \epsilon_i \quad \text{for every } i, \tag{1}$$

within a specified compute budget, where $\epsilon_i \geq 0$ are tolerances chosen before evaluating the joint run. The references measure separate transfer; they do not establish that these targets are jointly attainable. We report raw scores and compare learning curves at matched task exposure and total compute. The local analysis below concerns distillation losses; the experiments test how their changes relate to capability scores.

2.2 Dense supervision on student rollouts

There are M tasks with prompt distributions $\mathcal{D}_i$ and assigned frozen teachers π_{T_i}. The student and teachers share a vocabulary. For a task-i prompt x, the student generates a response y. At position u, the teacher scores the student-generated prefix $c = (x, y_{<u})$. Write $p_\theta(a \mid c) = \pi_\theta(a \mid c)$ and $q_i(a \mid c) = \pi_{T_i}(a \mid c)$.

We use the teacher top-k loss from MOPD (Ma et al., 2026). Let $S_i(c) = \mathrm{TopK}_k(q_i(\cdot \mid c))$. The position loss is

$$\ell_i^{(k)}(\theta; c) = \sum_{a \in S_i(c)} \left[p_\theta(a \mid c) \log \frac{p_\theta(a \mid c)}{q_i(a \mid c)} - p_\theta(a \mid c) + q_i(a \mid c) \right]. \tag{2}$$

The probabilities retain their full-vocabulary normalization. The $-p_\theta + q_i$ correction makes the loss smallest when student and teacher probabilities agree on the retained tokens. Both the teacher scores and the generated prefixes are held fixed during the update. This objective uses all retained token contributions at each observed position. For finite k, its gradient is that of the specified surrogate; it need not equal the full-vocabulary reverse-KL gradient.

Let $\mathcal{R}_i^t$ be the distribution of prompts and student responses available for task i at checkpoint θ_t, conditional on training history. We average positions within responses and responses within tasks:

$$L_i^t(\theta) = \mathbb{E}_{(x,y)\sim\mathcal{R}_i^t} \left[\frac{1}{|y|} \sum_{u=1}^{|y|} \ell_i^{(k)}(\theta; x, y_{<u}) \right]. \tag{3}$$

The distribution $\mathcal{R}_i^t$ is held fixed when differentiating. Positive weights w_i, fixed before training and summing to one, define

$$F_t(\theta) = \sum_{i=1}^{M} w_i L_i^t(\theta). \tag{4}$$

Thus $\mu_i = \nabla L_i^t(\theta_t)$ and $\mu_w = \sum_i w_i \mu_i = \nabla F_t(\theta_t)$ are the local mean gradients. Our main comparison uses equal weights.

Updating the student also changes the distribution of generated states. Adding and subtracting the loss on the previous distribution gives

$$\begin{aligned} L_i^{t+1}(\theta_{t+1}) - L_i^t(\theta_t) = {} & L_i^t(\theta_{t+1}) - L_i^t(\theta_t) \\ & + L_i^{t+1}(\theta_{t+1}) - L_i^t(\theta_{t+1}). \end{aligned} \tag{5}$$

The first difference measures optimization on the previous states; the second measures the change in the sampling distribution, including the student's new rollouts. We therefore evaluate loss on fixed reference prefixes separately from loss on fresh student responses. Neither loss alone establishes the capability target in Equation 1.

2.3 AVERAGING TASK GRADIENTS BEFORE COMBINING THEM

A micro-batch contains b prompts from one task. An optimizer step processes G micro-batches, with task i receiving m_i of them. The counts sum to G and satisfy $m_{\min} \leq m_i \leq m_{\max}$. We use $m_{\min} \geq 2$ to estimate each task's variability from the same step.

Let $g_{i,s}$ be the gradient of the response-mean loss in micro-batch s of task i. Fixed-weight comparisons accumulate

$$A = \sum_{i=1}^{M} w_i \underbrace{\frac{1}{m_i} \sum_{s=1}^{m_i} g_{i,s}}_{\text{task mean}} . \tag{6}$$

Implementation only requires multiplying each micro-batch loss by w_i/m_i. For example, an equally weighted task contributes one quarter of the combined mean whether it receives two or six micro-batches; six micro-batches give a more precise estimate of that mean.

Counts are chosen from history $\mathcal{H}_t$ before drawing the step's data. If each fresh task-i micro-batch has conditional mean μ_i, then $\mathbb{E}[A \mid \mathcal{H}_t, m] = \mu_w$. Changing counts preserves this mean gradient. It can still change AdamW's expected update through its moments and clipping. Response averaging removes the direct increase in task weight caused by longer responses, one source of the length imbalance studied by Open-MOPD (Gao et al., 2026).

3 WHEN CAN A SHARED UPDATE MAKE JOINT PROGRESS?

3.1 HOW TEACHER INFORMATION PRODUCES A TASK GRADIENT

An ideal reference distinguishes the teacher's target from the implemented update. If a sequence-level teacher is exactly $T_i^*(y \mid x) \propto \pi_0(y \mid x) \exp(R_i(x,y)/\beta_i)$, minimizing full-sequence reverse KL is equivalent to maximizing expected reward with KL regularization of strength β_i, up to a positive scale and an additive constant. Appendix A.5 derives the identity and its temperature-dependent mixture weights. It requires an exact teacher for the stated reward and reference. It does not make our fixed-prefix, response-mean, top-k update a full-sequence RL gradient.

For the implemented loss, let ν_i^t be the prefix distribution induced by sampling $(x,y) \sim \mathcal{R}_i^t$ and then a position uniformly within y. At θ_t, define

$$s_\theta(c,a) = \nabla_\theta \log p_\theta(a \mid c), \qquad z_i(c,a) = \mathbf{1}_{\{a \in S_i(c)\}} \log \frac{q_i(a \mid c)}{p_\theta(a \mid c)}. \tag{7}$$

Differentiating Equation 2 and using the full-vocabulary score identity gives

$$\begin{aligned} -\mu_i &= \mathbb{E}_{\nu_i^t, p_\theta}[z_i s_\theta] = \mathbb{E}_{\nu_i^t, p_\theta}[(z_i - \mathbb{E}[z_i \mid c]) s_\theta], \\ \|\mu_i\|_2 &\leq \sqrt{\mathbb{E}_{\nu_i^t} \operatorname{Var}_{p_\theta}(z_i \mid c)} \sqrt{\mathbb{E}_{\nu_i^t, p_\theta} \|s_\theta\|_2^2}. \end{aligned} \tag{8}$$

Teacher differences produce an update through their correlation with the student's score on visited prefixes. A large signal variance does not imply a large or useful gradient, and this variance is not micro-batch gradient covariance. Centering applies to the complete masked signal z_i: subtracting a baseline only inside $S_i(c)$ generally changes the update, since $\sum_{a \in S_i(c)} p_\theta s_\theta = \nabla_\theta p_\theta(S_i(c) \mid c)$ need not vanish. Appendix A.6 gives the proof and teacher-error qualification. These relations identify how prefix coverage, teacher probabilities, and retained support determine the mean gradient that the following analysis combines.

3.2 EXPECTED PROGRESS UNDER THE CHOSEN TASK MIXTURE

Condition on the training state, freeze the rollout distributions in L_i^t, and choose counts before sampling. Let $\mu_i = \nabla L_i^t(\theta_t)$ and $\Sigma_i = \operatorname{Cov}(g_{i,s})$. For the following local analysis, micro-batches are independent and have finite second moments. The diagnostic experiment uses independent draws; the training streams require the finite-population qualification in Appendix A.4.

Let D be a positive diagonal scaling fixed before the step. For SGD, $D = I$; for Adam, we use $D = \mathrm{diag}((\sqrt{\widehat{v}_{\mathrm{pre}}} + \epsilon)^{-1})$, where $\widehat{v}_{\mathrm{pre}}$ is the pre-step bias-corrected second moment. For the idealized update $\theta' = \theta_t - \eta DA$, define

$$\mu_w = \sum_j w_j \mu_j, \qquad K_{ij} = \mu_i^\top D \mu_j, \qquad a_i = (Kw)_i = \mu_i^\top D \mu_w. \tag{9}$$

The expected first-order change in task i's loss is $-\eta a_i$. Thus all $a_i > 0$ means that the chosen mean update is a common descent direction. A small positive a_i leaves little margin against sampling noise. If $a_i \le 0$, changing counts at fixed weights cannot produce strictly positive expected first-order progress for that task.

This last statement concerns this mixture at this checkpoint. It does not rule out another mixture, another update direction, or later recovery. Pairwise negative gradient cosines do not determine the signs of Kw. The relevant interaction uses one D; cosines between $D\mu_i$ and $D\mu_j$ instead use D^2. Actual AdamW includes momentum, an updated second moment, clipping, and weight decay. Its branch updates must be measured separately from this frozen-scaling approximation.

3.3 Common descent can still be slow

The signs of a_i do not specify the rate of progress. Consider exact gradients, $D = I$, equal weights, and

$$L_1(x, y) = \tfrac{1}{2}x^2, \qquad L_2(x, y) = \tfrac{\gamma}{2}y^2, \qquad 0 < \gamma < 1. \tag{10}$$

The task gradients are orthogonal everywhere. From $x(0) = y(0) = 1$, gradient flow for $(L_1 + L_2)/2$ gives

$$\frac{L_1(t)}{L_1(0)} = e^{-t}, \qquad \frac{L_2(t)}{L_2(0)} = e^{-\gamma t}. \tag{11}$$

Both tasks improve at every finite time, with no sampling uncertainty, yet their decay-rate ratio is $1/\gamma$. Figure 1(a) uses $\gamma = 0.1$. More generally, near a common matching solution of fixed-prefix losses, the local quadratic model has $\dot{e} = -DH(w)e$, where $H(w) = \sum_i w_i H_i$. For the corrected top-k loss, each matching-point H_i is the score second moment on the retained support, not generally the full Fisher matrix. Appendix A.7 states the assumptions.

This example establishes a possible cause of slow progress, not a convergence model for the full MOPD run. Small raw gradient inner products also do not establish improved angles: shrinking opposed gradients have inner products tending to zero with cosine -1, while nonzero orthogonal gradients already have zero inner product. Changing the task weights can change deterministic rates; changing counts m_i at fixed weights preserves the local mean field. The separable minimum-time weight calculation in Appendix A.8 therefore does not justify GPAS's sample allocation.

3.4 Sampling uncertainty relative to task progress

Let $X_i = \mu_i^\top DA$ be the sampled first-order progress. Its variance is

$$v_i(m) = \mathrm{Var}(X_i) = \sum_j \frac{w_j^2}{m_j} \mu_i^\top D \Sigma_j D \mu_i. \tag{12}$$

For $a_i > 0$, the one-sided Chebyshev–Cantelli inequality gives

$$\Pr(X_i \le 0) \le \frac{v_i(m)}{v_i(m) + a_i^2}, \qquad \Pr(\exists i : X_i \le 0) \le \min\left\{1, \sum_i \frac{v_i(m)}{v_i(m) + a_i^2}\right\}, \tag{13}$$

where the joint bound requires every $a_i > 0$. Reliability therefore depends on noise relative to the progress margin, not on raw noise alone. These bounds require no Gaussian assumption, but can be loose. Replacing population quantities with noisy estimates does not produce a statistical certificate. Appendix A.9 proves the bounds; Appendices A.10 and A.13 give a sufficient sampling budget and the additional curvature conditions for finite-step descent. Directional sampling tests and stochastic multiobjective sampling have established precedents (Bollapragada et al., 2018; Zhao et al., 2024); our empirical question is whether these quantities predict MOPD regressions and useful interventions.

1. Choose feasible integer counts from the previous noise and time averages; use Uniform on the first step. Read the pre-step scaling D.
2. For each task, generate m_i micro-batches and obtain its teacher's top-k scores on the student responses.
3. Compute the dense loss, accumulate unweighted task gradients, record their variation under D, and add the task mean to A with weight w_i.
4. Update the noise and time averages, then apply the usual optimizer step with gradient A.

Algorithm 1: One training step with GPAS.

3.5 GPAS CONTROLS A CONSERVATIVE VARIANCE BOUND

GPAS measures total gradient variation after optimizer scaling:

$$e_i = \mathbb{E}\|D(g_i - \mu_i)\|_2^2, \qquad V(m) = \mathbb{E}\|D(A - \mu_w)\|_2^2 = \sum_i \frac{w_i^2 e_i}{m_i}. \tag{14}$$

It controls each directional variance through

$$v_i(m) \leq \|\mu_i\|_2^2 V(m). \tag{15}$$

This is a conservative surrogate: fluctuations orthogonal to a task's gradient contribute to V without affecting its first-order progress. Appendix A.12 gives aligned task means for which trace-optimal allocation has worse directional reliability than another allocation. GPAS is therefore an intervention to test, not an optimal rule for joint-progress probability.

Ignoring integer bounds, minimizing V at $\sum_i m_i = G$ yields

$$m_i \propto w_i \sqrt{e_i}. \tag{16}$$

This is classical Neyman allocation (Neyman, 1934; Cochran, 1977) in the optimizer's coordinates. The four-task implementation enumerates all feasible integer counts and minimizes the estimated V exactly over that set; Appendix A.11 gives the derivation. Under the study's bounds, $m_i \leq 8$ versus Uniform's $m_i = 4$ implies $V_{\text{Uniform}}/V(m) \leq 2$ when the denominator is positive. This limits variance reduction under the local model, not training speed.

3.6 ONLINE ESTIMATES AND MEASURED COMPUTATION

GPAS reuses the unweighted micro-batch gradients. With task mean $\bar{g}_i$,

$$\widehat{e}_i = \frac{1}{m_i - 1} \sum_{s=1}^{m_i} \|D(g_{i,s} - \bar{g}_i)\|_2^2. \tag{17}$$

The next step uses the moving average $\widetilde{e}_i \leftarrow \rho \widetilde{e}_i + (1-\rho)\widehat{e}_i$. A running mean and scalar squared-deviation sum avoid retaining all micro-batch gradients; one parameter-sized mean buffer is reused across task blocks. The first step uses Uniform. We measure the additional reductions, memory, and time rather than assuming negligible overhead.

For marginal micro-batch time τ_i and fixed step time C, the cost-aware mode minimizes

$$J_C(m) = T(m)V(m) = \left(C + \sum_i m_i \tau_i\right) \sum_i \frac{w_i^2 e_i}{m_i}. \tag{18}$$

This criterion measures time times variance; it does not generally equal learning progress per second. The continuous rule without count bounds at $C = 0$ is $m_i \propto w_i \sqrt{e_i/\tau_i}$, and the finite implementation enumerates feasible counts. The timing model is fitted to measured task times and checked against end-to-end GPU hours. Directional allocation is a conditional extension in Appendix A.14, to be investigated only if trace noise fails to predict the uncertainty that affects task progress.

4 Experimental Protocol

The protocol first tests individual transfer, then distinguishes weak own-task signals, effects of combining tasks, sampling failures, and reliable but slow local progress at common checkpoints. It finally evaluates prediction, capability, and computation. End-to-end measurements are not yet available; the results below specify the evidence to collect.

4.1 A four-domain distillation setting

We study mathematics, code, instruction following, and science with a Qwen3-1.7B student in non-thinking mode. Mathematics and instruction following use domain-specific Qwen3-1.7B RL teachers. Code and science are routed to the same frozen Qwen3-4B model through separate task endpoints. This setting combines specialized feedback and a larger generalist teacher in one student.

The default loss is Equation 2 with teacher $k = 64$, and each task has weight $w_i = 1/4$. Each step processes $G = 16$ micro-batches of $b = 4$ responses, with $2 \leq m_i \leq 8$. Uniform uses four micro-batches per task. Runs last 500 steps; each task supplies a shuffled stream of 16,000 unique prompts, and each response is capped at 4,096 tokens. There is one optimizer update per fresh rollout batch. Noise and time estimates use EMA decay 0.9. Runs use AdamW and shared task prompt orders within each paired seed. Uniform and GPAS (per step) use three paired training seeds; the other seven joint configurations use one exploratory seed each.

One 96-GB GPU trains the student, and one 48-GB GPU serves student rollouts and teacher scoring. The two generalist task endpoints share their teacher weights in memory. Appendix A.1 records the configuration. **[TODO: Complete the exact checkpoint and data revisions, optimizer settings, software versions, and measured resource use.]**

4.2 Individual transfer and joint training

Four single-task OPD references use the same initial student, assigned teacher, and default loss with weight one. Each initially uses one seed, 500 updates, and 16 micro-batches per update. Its 32,000 responses require two passes through the 16,000-prompt pool, reshuffling only after exhaustion. Joint runs do not repeat prompts. We report unique prompts, exposure, and compute, and compare learning curves at matched domain exposure and total compute. The reference's final score is an observed separately acquired capability, neither an upper bound nor evidence of joint attainability.

The fixed-objective comparisons change only the assignment of the 16 micro-batches:

- **Uniform**: four micro-batches per task.
- **GPAS (per step)**: counts from smoothed preconditioned gradient noise.
- **GPAS (cost-aware)**: counts from the time–variance criterion in Equation 18.
- **Unpreconditioned allocation**: the same variance rule with $D = I$ for count selection; AdamW still performs the update.
- **D^3 signal, fixed weights**: counts from D^3-MOPD's remaining-gap and descent-velocity signal, combined with our fixed task-mean weights.

Two recipe comparisons change the objectives. **D^3-MOPD scheduling** averages responses across the step, giving effective task weights m_i/G. **Open-MOPD** ($K = 1$) uses its student top-16 dense loss, token-share balancing, and gap-aware weighting. Appendix A.3 specifies these adaptations to our models, data, and budget. **TA-OPD + Uniform** and **TA-OPD + GPAS** use the tail-aware loss of Huang & Wei (2026) at $k = 64$ to test loss sensitivity. These nine joint configurations accompany the four single-task references. Only the repeated Uniform–GPAS comparison supports claims about training-seed stability.

4.3 Mean strength, interference, and precision

At Uniform steps 50, 250, and 450, each diagnostic branch restores the same complete student, AdamW, scheduler, data, and random state and makes one update. We compare each task alone, the Uniform joint update, GPAS, and a more precise equal-weight joint estimate using 32 micro-batches per task. Single-task branches use 16 micro-batches of their task with weight one; Uniform and GPAS retain $G = 16$. The precise branch is an expensive diagnostic, and its cost is reported separately.

At each checkpoint, calibration, update-trial, and evaluation samples are drawn independently from the same declared diagnostic distribution for each task, using the fixed checkpoint rollout policy. These distributions are held out from training and benchmarks. For these tests, L_i^t and μ_i refer to the diagnostic distribution; applicability to the training distribution is evaluated empirically. Draws and responses are not reused across roles; independent draws may select the same prompt. Calibration uses 32 independent micro-batches per task to estimate means, trace noise, transfer margins a_i (Equation 9), and directional variances (Equation 12). We split calibration into two independent halves and use cross-fitted mean products for K and a_i, avoiding the upward bias from squaring the same noisy mean estimate. We compare the self-progress term K_{ii}, its weighted contribution $w_i K_{ii}$, and the combined margin a_i, with uncertainty, alongside the existing single-task and precise joint branches. This distinguishes a weak own-task signal from a reduction after task combination. Calibration estimates alone select diagnostic GPAS counts. We run 20 independent update trials per branch and evaluate every branch on the same independent, fixed reference prefixes for all four domains. Diagnostic samples are independent with replacement; joint training uses finite streams without replacement. Their covariance assumptions are distinct.

For each trial we use independent evaluation gradients to estimate the first-order projection under frozen pre-step D, and record the fixed-prefix loss change after an actual AdamW update. A separate equal-allocation sweep uses $m_i \in \{2, 4, 8, 16, 32\}$ for every task, so G ranges from 8 to 128. This local sweep is exempt from the main count cap and does not test the end-to-end effect of changing update frequency under a fixed compute budget.

We classify a harmful mean interaction only when an independently evaluated margin estimate is negative beyond its sampling uncertainty. A precise AdamW update can increase loss because of momentum or finite-step effects even with a positive frozen-D margin; those cases test the approximation. Positive margins with frequent sampled sign reversals identify a candidate sampling failure. Margins whose uncertainty includes zero remain inconclusive. We also classify reliable but slow local progress when loss decreases consistently yet its absolute and relative changes fall below prespecified thresholds. Relative changes use the pre-update loss only when it exceeds a prespecified floor. A single checkpoint does not identify curvature, capacity, or a convergence rate. The analytic examples establish possibilities; measured loss changes test whether those cases occur here. Total trace noise is assessed as a proxy for directional uncertainty; a directional allocation is considered only as an offline diagnostic if that proxy fails.

4.4 Prediction before intervention

We specify the diagnostic rule, regression and slow-progress thresholds, confidence procedure, and comparison metrics using step 50 of the first Uniform seed, before examining later diagnostic outcomes. The rule uses self-progress, combined margin magnitude and sign, and v_i/a_i^2 for positive margins to predict regressions, reliable but small loss decreases, and the effect of increased precision. We test whether weak own-task progress remains slow after precise sampling and whether combination reduces otherwise useful progress. Steps 250 and 450 and the two other training seeds evaluate these predictions without retuning the rule. We compare predictions from directional uncertainty with those from trace noise alone, and report predictive errors as well as successful interventions. Plug-in estimates of the probability bound are diagnostic quantities, not finite-sample certificates. **[TODO: Record the rule and analysis specification before evaluating the held-out checkpoints and seeds.]**

4.5 Capability, state distribution, and computation

Separate measurements. We report MATH-500 greedy pass@1, a fixed LiveCodeBench pass@1 slice, IFBench strict accuracy, and GPQA-Diamond average@4. The initial student and assigned teachers use the same benchmark protocols. Raw per-domain scores remain primary. Reference-normalized gains are reported only when the single-task gain exceeds a prespecified positive denominator floor; otherwise we report raw gains. Every 50 steps, all methods are evaluated with the

[TODO: Plot single-task and joint capability gains per domain against domain exposure and total GPU hours. Mark prompt repetition in single-task references and show raw scores. Measurements are not yet available.]

Figure 2: Individual transfer and joint training under explicit exposure and compute comparisons. A difference does not identify its cause.

common teacher top-64 loss and equal task weights on 128 fixed held-out prompts per task. We keep two distinct loss measurements: a fixed bank of initial-student reference prefixes, and fresh prefixes generated by the current student. Common-checkpoint diagnostics additionally use an independent bank generated at that checkpoint and held fixed across its branches. We retain each checkpoint's fresh reference bank and teacher scores, then evaluate the next checkpoint on both the previous and new banks. These paired measurements give the empirical counterpart of Equation 5 on the held-out prompt distribution over 50-step intervals, with sampling uncertainty reported separately. The initial bank provides a common fixed-loss target; its difference from fresh-policy loss alone is not that decomposition. Neither loss is treated as a capability score.

The main capability table uses step 500 for every run, without selecting a different checkpoint for each domain. For the repeated core comparison, we report all three seed outcomes and their mean and spread. Paired prompt-level bootstrap intervals describe evaluation uncertainty separately; they do not replace training-seed variation. Single-task references initially have one seed, which limits claims about their stability. **[TODO: Insert benchmark versions, code evaluation date range, complete decoding settings, the reference-gain denominator floor, and capability tolerances ϵ_i before inspecting results.]**

Measured efficiency. We plot fixed-reference loss, fresh-policy loss, and capability separately against task exposure, optimizer steps, and GPU hours. GPU hours include all time on the two reserved devices, including waiting, synchronization, statistics, and checkpointing. Diagnostic branches and evaluation have separate compute totals. Within the fixed-weight default-loss group, we report time to Uniform's mean final fixed-reference loss, interpolating between adjacent evaluations and marking unreached targets. This is a loss-based efficiency measure; per-domain capability and the worst-domain change accompany it.

Allocation traces. Using existing top-64 outputs, we additionally log student and teacher retained mass and the response-mean conditional variance of the centered masked log ratio. This scalar describes the teacher signal in the specified surrogate; it is not gradient covariance or a certificate of useful signal. It requires no full-vocabulary teacher output or additional gradient buffer. The training loop logs task loss, its recent rate of change, response length, trace noise with and without preconditioning, assigned counts, task exposure, and time per micro-batch. Diagnostic checkpoints add progress margins and directional noise. We test whether these measurements explain observed regressions and intervention effects. The cost-aware criterion measures time times update variance; its relationship to learning progress is evaluated empirically. All statistics collection costs enter the time and memory report.

5 Results

5.1 Separately acquired and jointly acquired capabilities

Figure 2 tests whether the chosen student acquires each capability separately and whether joint training reaches comparable gains. A task that does not improve alone does not establish an integration

[TODO: For each checkpoint and domain, show cross-fitted self and combined progress with uncertainty, masked teacher signal statistics, directional and trace noise, and absolute and relative fixed-prefix loss changes for single-task, Uniform, GPAS, and precise-gradient branches. Add the local batch-size sweep. All panels await measurements.]

Figure 3: Common-checkpoint interventions compare own-task and combined progress, sampling sensitivity, and reliable but small loss decreases, while testing the frozen-D approximation against AdamW updates.

Table 1: Four-domain capability at step 500. Blank cells await measurements. Uniform and GPAS use three seeds; all other joint configurations and each single-task reference initially use one. The single-task row contains four separate students, each scored in its training domain, not one joint model.

Method	Math	Code	IF	Science	Mean
Initial student					
Assigned teachers					
Single-task references					
Uniform					
GPAS (per step)					
GPAS (cost-aware)					
Unpreconditioned allocation					
D^3 signal, fixed weights					
D^3-MOPD scheduling					
Open-MOPD ($K = 1$)					
TA-OPD + Uniform					
TA-OPD + GPAS					

failure. A joint-versus-single-task gap requires the common-checkpoint tests before it can be attributed to interference or sampling. **[TODO: Report raw gains, uncertainty, matched-exposure comparisons, and any tasks for which individual transfer is weak or inconclusive.]**

5.2 Mean strength, slow progress, and sampling failures

[TODO: Report which estimated mean margins are positive, negative, or inconclusive; whether weak own-task progress or task combination reduces useful progress; and whether increased precision reduces sign failures or leaves reliable but slow local progress. Report absolute and relative loss changes and descriptive teacher-signal statistics. Include uncertainty in the precise estimate and cases where finite-step AdamW behavior differs from the local projection. Then report predictive performance on held-out checkpoints and seeds, comparing directional uncertainty with trace noise. Do not infer curvature, capacity limits, or convergence rates from a single checkpoint; the analytical examples alone establish none of these for the trained models.]

5.3 Capability and computation after intervention

Table 1 and Figure 4 test whether the diagnosed local effect leads to useful end-to-end changes. **[TODO: Report every per-domain change, the worst-domain change, seed variation, measured**

[TODO: Plot fixed-reference loss, fresh-policy loss, and per-domain capability in separate panels against steps and measured GPU hours. Show three-seed variation for Uniform and GPAS; label secondary comparisons as single-seed exploratory runs. Measurements are not yet available.]

Figure 4: Training progress, changes in visited states, and capability evaluated separately under the same compute accounting.

[TODO: Align task counts, trace noise, loss, and response length over training; mark the diagnostic checkpoints and their progress margins. Keep preconditioned and unpreconditioned noise on separate scales. Measurements are not yet available.]

Figure 5: Allocation traces interpreted using the measured progress margins and intervention outcomes.

GPU hours, and time to the fixed-reference loss target. State whether capability and the two loss measurements agree. Report negative or null results alongside improvements.]

5.4 Allocation, cost, and dense-loss sensitivity

The unpreconditioned and fixed-weight D^3 comparisons test the role of optimizer scaling and the choice of allocation signal. The cost-aware run tests whether minimizing time times variance improves measured efficiency. The TA-OPD pair tests sensitivity to the dense loss. **[TODO: Relate allocation changes to measured directional uncertainty and outcomes. Report statistics overhead, fitted and observed task costs, and capability changes under the second loss. Keep conclusions from single-seed comparisons exploratory.]**

6 Related Work

Capability integration in multi-teacher distillation. MOPD combines domain teachers through supervision on student-generated responses (Ma et al., 2026). Open-MOPD measures integration gaps against separately distilled students and addresses response-token imbalance, uneven progress, and stale student-dependent rewards (Gao et al., 2026). D^3-MOPD adapts domain sampling using the remaining teacher gap and its descent velocity (Sun et al., 2026). CaMOPD studies counteracting recovery and preservation gradients and combines alternating training with gap-based selection (Chen et al., 2026). Our analysis relates the implemented teacher signal to task gradients and distinguishes mean progress from uncertainty in its estimate. Our proposed checkpoint interventions test whether increased precision improves joint progress under fixed weights.

Dense losses and token estimators. MOPD includes both sampled-token policy gradients and a corrected reverse-KL loss on the teacher's top-k tokens (Ma et al., 2026). Open-MOPD uses student top-k candidates with probability-weighted log-ratio advantages; its released dense implementation computes candidate-wise losses before summing across the retained set (Gao et al., 2026;

BytedTsinghua-SIA, 2026). EMA-PG combines an exact retained-set contribution with a sampled tail (Zhang & Ba, 2026), TA-OPD includes the probability mass outside the teacher's top-k set (Huang & Wei, 2026), and vOPD uses a control-variate baseline for sampled-token distillation (Oh et al., 2026). We build on existing dense losses and apply GPAS to the micro-batch gradients they produce.

On-policy learning dynamics. Teacher–student compatibility and the capabilities supplied by the teacher shape OPD outcomes (Li et al., 2026). Rethinking OPD II finds slow alignment even when training states are held fixed in its studied settings (Fu et al., 2026). This motivates examining optimization separately from state coverage, but does not identify gradient noise as the cause. We therefore distinguish fixed-reference loss, fresh-policy loss, and downstream capability in the evaluation protocol.

Task weighting and stochastic optimization. Multi-task learning adapts loss weights and gradient directions (Chen et al., 2018; Sener & Koltun, 2019; Yu et al., 2020); language-model training also adapts data mixtures from learning progress (Albalak et al., 2023; Wang et al., 2025). Directional sampling tests already distinguish descent reliability from norm accuracy (Bollapragada et al., 2018), and adaptive sampling has been developed for stochastic multiobjective optimization (Zhao et al., 2024). Our local analysis applies these established ideas to domain progress in MOPD. GPAS minimizes total update variance, following Neyman allocation and its unequal-cost extension (Neyman, 1934; Cochran, 1977). The interaction between non-uniform sampling and adaptive optimizers (Lahire, 2023) motivates measuring variance after the optimizer's coordinate scaling. We show why minimizing this trace variance need not maximize each domain's directional reliability.

7 Discussion and Conclusion

Teacher information produces an update through its correlation with the student's score on visited prefixes and retained tokens. The local analysis then distinguishes the direction and rate of mean progress from sampling uncertainty. The zero-conflict example shows that exact common descent can still have very different task learning rates. When a task's progress margin is nonpositive, changing sample counts at fixed weights cannot make its expected first-order progress positive under the frozen-preconditioner model. For positive margins, additional samples can reduce uncertainty; this does not ensure sufficient progress within the training budget. GPAS allocates samples to reduce total update variance, which bounds directional variance conservatively. Directional variance determines a more specific bound for each task; the two allocation criteria need not agree.

The proposed study tests these quantities against updates from identical checkpoints and against complete training runs. Separate-distillation references assess individual transfer, while fixed-state loss, fresh-policy loss, and capability evaluation measure different aspects of training. **[TODO: Report which diagnostics predicted observed failures, when GPAS improved capability or compute efficiency, and which failures remained.]**

These local conditions do not establish joint capability attainability or global convergence. Successful training may temporarily increase a task loss, and a local conflict does not prove a capacity limit. The empirical scope is one student size, four domains, and one hardware layout; the repeated core comparison estimates training variation within that scope. Finite-sample diagnostic error, changing optimizer states, and changing rollout distributions must be assessed before using the local bounds to make training decisions.

AI use statement

Generative AI tools assisted with language editing, manuscript organization, mathematical analysis, and experimental design. The authors remain responsible for reviewing the text, claims, proofs, code, and experimental results before submission.

Ethics statement

The study evaluates optimization methods for existing language models with reasoning and instruction-following datasets and automated verifiers. Model and dataset biases can affect capability scores. The final artifact will record data provenance, invalid outputs, and verifier failures to support careful interpretation.

Reproducibility statement

Section 4 specifies separate-distillation references, checkpoint interventions, diagnostic data separation, allocation comparisons, and resource accounting. Appendices A.5–A.7 derive the ideal reference, implemented teacher signal, and local rate example; Appendix A.9 gives the reliability bounds. The current artifact includes manuscript sources, a reproducible analytical teaser, and earlier synthetic calculations; the end-to-end MOPD trainer and measured results remain to be added. **[TODO: Before submission, add the frozen run configuration, model and tokenizer revisions, fixed task weights, prompt streams, optimizer and allocation states, diagnostic data splits, reference prefixes, independent trial gradients, micro-batch counts, noise and timing logs, per-seed and per-prompt evaluation outputs, and source data for every empirical figure and table.]**

References

Alon Albalak, Liangming Pan, Colin Raffel, and William Yang Wang. Efficient online data mixing for language model pre-training, 2023. URL `https://arxiv.org/abs/2312.02406`.

Raghu Bollapragada, Richard Byrd, and Jorge Nocedal. Adaptive sampling strategies for stochastic optimization. *SIAM Journal on Optimization*, 28(4):3312–3343, 2018. doi: 10.1137/17M1154679. URL `https://epubs.siam.org/doi/10.1137/17M1154679`.

BytedTsinghua-SIA. Open-MOPD: Dense distillation implementation. GitHub, 2026. URL `https://github.com/BytedTsinghua-SIA/Open-MOPD`. Revision 4809a96cf85a, actor and dense-loss implementation.

Tianlei Chen, Jiao Ou, Ziyuan Liu, Ruiming Tang, Jian Liang, and Han Li. Counteraction-aware multi-teacher on-policy distillation for general capability recovery with domain preservation, 2026. URL `https://arxiv.org/abs/2605.27115`.

Zhao Chen, Vijay Badrinarayanan, Chen-Yu Lee, and Andrew Rabinovich. Gradnorm: Gradient normalization for adaptive loss balancing in deep multitask networks. In *Proceedings of the 35th International Conference on Machine Learning*, pp. 794–803, 2018. URL `https://proceedings.mlr.press/v80/chen18a.html`.

William G. Cochran. *Sampling Techniques*. Wiley, 3 edition, 1977.

Zixuan Fu, Bingxiang He, Yuxin Zuo, Haohuan Huang, Jinqian Zhang, Ruhang Xiao, Cheng Qian, Qinyu Luo, Huan ang Gao, Yudong Wang, Zhiyuan Liu, Ning Ding, and Chaojun Xiao. Rethinking on-policy distillation of large language models II: One training example, 2026. URL `https://arxiv.org/abs/2609.04172`.

Huan-ang Gao, Haohan Chi, Yong Yan, Shiyuan Feng, Hanlin Wu, Zheng Jiang, Bingxiang He, Wei-Ying Ma, Ya-Qin Zhang, and Hao Zhou. Open-MOPD: Diagnosing and fixing capability imbalance in multi-teacher on-policy distillation, 2026. URL `https://arxiv.org/abs/2608.19098`.

Huipeng Huang and Hongxin Wei. Tail-aware top-k on-policy distillation, 2026. URL `https://arxiv.org/abs/2608.14728`.

Thibault Lahire. Non-uniform sampling and adaptive optimizers in deep learning. OPT 2023: Optimization for Machine Learning, 2023. URL `https://opt-ml.org/papers/2023/paper20.pdf`.

Yaxuan Li, Yuxin Zuo, Bingxiang He, Jinqian Zhang, Chaojun Xiao, Cheng Qian, Tianyu Yu, Huan ang Gao, Wenkai Yang, Zhiyuan Liu, and Ning Ding. Rethinking on-policy distillation of large language models: Phenomenology, mechanism, and recipe, 2026. URL `https://arxiv.org/abs/2604.13016`.

Jiang Liu, Sudhanshu Ranjan, Prakamya Mishra, Yonatan Dukler, Gowtham Ramesh, Jialian Wu, Ximeng Sun, Wen Xie, Chaojun Hou, Vikram Appia, Zhenyu Gu, Zicheng Liu, and Emad Barsoum. Instella-MoE technical report, 2026. URL https://arxiv.org/abs/2609.00791.

Wenhan Ma, Jianyu Wei, Liang Zhao, Hailin Zhang, Bangjun Xiao, Lei Li, Qibin Yang, Bofei Gao, Yudong Wang, Rang Li, Jinhao Dong, Zhifang Sui, and Fuli Luo. MOPD: Multi-teacher on-policy distillation for capability integration in llm post-training, 2026. URL https://arxiv.org/abs/2606.30406.

Jerzy Neyman. On the two different aspects of the representative method: The method of stratified sampling and the method of purposive selection. *Journal of the Royal Statistical Society*, 97(4): 558–625, 1934. doi: 10.2307/2342192.

Minjae Oh, Sangjun Song, Gyubin Choi, Yunho Choi, and Yohan Jo. KL for a KL: On-policy distillation with control variate baseline, 2026. URL https://arxiv.org/abs/2605.07865.

Qwen Team. Qwen3 technical report, 2025. URL https://arxiv.org/abs/2505.09388.

Ozan Sener and Vladlen Koltun. Multi-task learning as multi-objective optimization, 2019. URL https://arxiv.org/abs/1810.04650.

Zechen Sun, Zhiwei Zhang, Fei Zhao, Juntao Li, Mu Chuan, Huayu Deng, Guojian Zhan, Wenliang Chen, Yao Hu, and Min Zhang. D^3-MOPD: Adaptive dynamic domain scheduling for efficient multi-teacher distillation, 2026. URL https://arxiv.org/abs/2608.24987.

Zhenting Wang, Guofeng Cui, Kun Wan, and Wentian Zhao. DUMP: Automated distribution-level curriculum learning for RL-based LLM post-training, 2025. URL https://arxiv.org/abs/2504.09710.

Tianhe Yu, Saurabh Kumar, Abhishek Gupta, Sergey Levine, Karol Hausman, and Chelsea Finn. Gradient surgery for multi-task learning, 2020. URL https://arxiv.org/abs/2001.06782.

Lunjun Zhang and Jimmy Ba. EMA policy gradient: Taming reinforcement learning for LLMs with EMA anchor and top-k KL, 2026. URL https://arxiv.org/abs/2602.04417.

Yong Zhao, Wang Chen, and Xinmin Yang. Adaptive sampling stochastic multigradient algorithm for stochastic multiobjective optimization. *Journal of Optimization Theory and Applications*, 200:215–241, 2024. doi: 10.1007/s10957-023-02334-w. URL https://link.springer.com/article/10.1007/s10957-023-02334-w.

A IMPLEMENTATION AND SUPPORTING DERIVATIONS

A.1 CONFIGURATION

Table A1: Configuration of the four-domain study. Entries marked TODO require the corresponding training or evaluation artifact.

Component	Configuration
Student	Qwen3-1.7B (Qwen Team, 2025), non-thinking mode; **[TODO: frozen model and tokenizer revisions]**
Teachers	Domain-specific Qwen3-1.7B RL checkpoints for math and instruction following; one shared Qwen3-4B weight set for code and science. All use the same vocabulary and tokenizer. **[TODO: exact checkpoint identifiers, RL training recipes, and scores]**
Training data	16,000 unique prompts per task. Joint runs consume shuffled streams without replacement; single-task references reshuffle only after exhausting the first cycle. **[TODO: datasets, versions, licenses, filters, and held-out split]**
Default loss	Teacher top-64 corrected reverse KL (Equation 2); full-vocabulary probabilities; position mean within response, response mean within task; $w_i = 1/4$
Rollout	One response per prompt and one update per fresh batch; maximum response length 4,096; truncated responses retained. **[TODO: temperature, EOS handling, and decoding settings]**
Step and budget	Joint runs: $b = 4$, $G = 16$, $2 \le m_i \le 8$, 500 updates and 32,000 attempted responses per run. Uniform and per-step GPAS: three training seeds; seven other configurations: one exploratory seed each. Four single-task references: one seed each, $G = 16$ on the active task and 500 updates
Optimizer	AdamW; **[TODO: learning rate and schedule, betas, epsilon, weight decay, clipping, precision, and memory configuration]**
GPAS	Pre-step bias-corrected Adam second moment for D; noise and time EMA decay 0.9; first step Uniform; exact feasible-integer enumeration for both per-step and cost-aware counts
Checkpoint diagnostics	All three Uniform seeds at updates 50, 250, and 450; seed-1 update 50 sets the prediction rule. Calibration: 32 independent micro-batches per domain split into two halves for cross-fitted margins; independent trial and evaluation data. Precise gradient: 32 micro-batches per domain; equal-count sweep: $m_i \in \{2, 4, 8, 16, 32\}$ outside the main run's count caps
Evaluation	128 held-out prompts per task every 50 steps; MATH-500, LiveCodeBench, IFBench, and GPQA-Diamond. **[TODO: versions, evaluation seeds, and decoding protocols]**
System	One 96 GB learner GPU and one 48 GB rollout/teacher GPU; the code and science endpoints share weights. **[TODO: GPU models, inference and training software revisions, peak memory, and GPU hours]**

A.2 DENSE LOSS IMPLEMENTATION AND ONLINE STATISTICS

At each student-generated prefix, the teacher returns the identifiers and full-vocabulary log probabilities of its top-64 tokens. The student computes its full-vocabulary log normalizer and gathers its probabilities at those identifiers. Autodifferentiation is applied to the whole expression in Equation 2, including the $-p_\theta$ term. The retained probabilities are not renormalized within the top-k set, and the loss does not use a sampled tail correction. The operation is dense over the retained set and uses sparse teacher output.

GPAS collects statistics from unweighted micro-batch gradients. For a task block, initialize its running mean $h = 0$ and scalar $S = 0$. After micro-batch s, form $\delta = g_{i,s} - h$, then update

$$h \leftarrow h + \delta/s, \qquad S \leftarrow S + \frac{s-1}{s}\|D\delta\|_2^2. \tag{19}$$

At the end of the block, $\widehat{e}_i = S/(m_i - 1)$ and $A \leftarrow A + w_i h$. The mean buffer is reused for the next task. The same recurrence with $D = I$ records unpreconditioned noise. Reductions use float32 accumulation. Gradients are unscaled before measurement if mixed-precision loss scaling is active, and clipping is applied to the assembled gradient A.

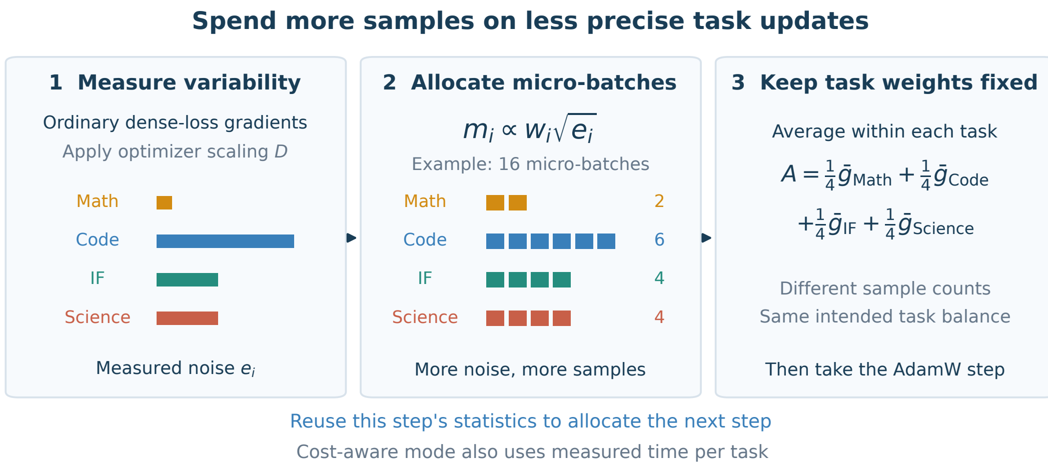

Figure 6: GPAS combines task means with fixed weights w while allocating counts m from observed gradient variation and, in its cost-aware mode, measured time. The pre-step scaling D defines the variance statistic. The profiles are schematic and do not report measured MOPD results or certify joint task progress.

The optimizer state is fixed during this collection. Before an Adam second moment is available, the first Uniform step uses $D = I$. Subsequent steps use the pre-step bias-corrected second moment. The first observation initializes each noise EMA; later observations use decay 0.9. The scheduler reads the completed step's averages before generating the next step's data.

For both allocation modes, enumerate $\mathcal{M} = \{m \in \mathbb{Z}^M : \sum_i m_i = G,\, m_{\min} \leq m_i \leq m_{\max}\}$ and evaluate the objective using the previous step's smoothed statistics. Per-step GPAS minimizes $\sum_i w_i^2 \tilde{e}_i / m_i$; the cost-aware mode minimizes its product with the estimated step time. Ties first minimize distance to Uniform and then follow lexicographic task order. If all noise estimates are zero, use Uniform. This finite search is exact for the estimated objective and stated integer constraints. Largest-remainder rounding of the continuous solution does not in general solve that integer problem. No current-step observations are used to select the counts of the same step.

A.3 Comparison recipes

D^3-MOPD scheduling. We use the remaining-gap and descent-velocity construction of Sun et al. (2026). Let $\bar{L}_i(t)$ be the smoothed task loss and $\bar{L}_i^0$ the mean of its first five observations. With $W = 10$, $R' \leq R = 3$ the number of available windows, and $\epsilon_{\mathrm{KL}} = 0.15$, the composite signal is

$$\begin{aligned} r_i(t) &= \bar{L}_i(t) / \max\{L_i^0, \epsilon_{\mathrm{KL}}\}, \\ \nu_i(t) &= \max\left\{0, \frac{1}{R'} \sum_{j=0}^{R'-1} \frac{\bar{L}_i(t-(j+1)W) - \bar{L}_i(t-jW)}{\max\{\bar{L}_i(t-(j+1)W), \epsilon_{\mathrm{KL}}\}}\right\}, \\ s_i(t) &= r_i(t)\nu_i(t). \end{aligned} \tag{20}$$

We retain the published max-normalization, softmax temperature 0.5, per-domain probability floor 0.10, and multiplicative batch jitter of amplitude 0.30. The scheduler refreshes every 10 steps, using the paper's history warmup and EMA window of 10. All KL denominators use the published floor of 0.15. The signal is computed from the default dense task loss in our pipeline. The resulting mixture is converted to micro-batch counts with the shared bounds $[2, 8]$. Thus this is a controlled adaptation to our loss scale, micro-batch unit, and budget.

The two D^3 rows use the same scheduling rule. The fixed-weight row assembles $A = \sum_i w_i \bar{g}_i$; the recipe-level row assembles $A = \sum_i (m_i/G)\bar{g}_i$. This makes the effect of task weighting visible alongside the comparison of scheduling signals.

Open-MOPD with one update per rollout. We follow the released student top-16 dense candidate loss, token-share balancing, and gap-aware weighting (Gao et al., 2026; BytedTsinghua-SIA, 2026). The released implementation multiplies candidate-wise detached advantages by candidate-wise student probability ratios and then sums over candidates. Each retained token therefore contributes its own gradient. In our setting, the initial task target is uniform, and each batch contains four micro-batches per task. The recipe's token and gap weights remain active. One optimizer step per rollout batch corresponds to $K = 1$, so there is no reuse-induced reward staleness. **[TODO: Record the exact configuration for gap smoothing, floors, and weight normalization from the integrated implementation.]**

Tail-aware supervision. TA-OPD (Huang & Wei, 2026) groups the omitted vocabulary into one tail event. For teacher support $S_i(c)$, write $p_{\text{tail}} = 1 - \sum_{a \in S_i(c)} p_\theta(a \mid c)$ and $q_{\text{tail}} = 1 - \sum_{a \in S_i(c)} q_i(a \mid c)$. Its position loss is

$$\ell_i^{\text{TA}} = \sum_{a \in S_i(c)} p_\theta(a \mid c) \log \frac{p_\theta(a \mid c)}{q_i(a \mid c)} + p_{\text{tail}} \log \frac{p_{\text{tail}}}{q_{\text{tail}}}. \tag{21}$$

Both runs use $k = 64$, the same response averaging and fixed task weights, and the same training budget as the core comparison. Only the allocation differs between TA-OPD + Uniform and TA-OPD + GPAS.

A.4 Frozen objectives, sampling, and state-distribution change

The task loss L_i^t in Equation 3 holds the student's step-t rollout distribution $\mathcal{R}_i^t$ fixed while varying θ. Under differentiation–expectation interchange, $\mu_i = \nabla L_i^t(\theta_t)$ and $\mu_w = \nabla F_t(\theta_t)$. Counts chosen from history before sampling preserve this weighted mean when the conditional task sampling law is unchanged. This is a sequence of frozen-distribution objectives; it does not differentiate through the policy that generated the prefixes.

Equation 5 separates optimization on the old states from the change in visited states. More explicitly, if $f_i(\theta; x, y)$ is the response-mean retained-support loss, its second term is

$$\int f_i(\theta_{t+1}; x, y)\, d(\mathcal{R}_i^{t+1} - \mathcal{R}_i^t)(x, y). \tag{22}$$

This term has no general sign. For example, lower moving-policy loss could reflect a shift toward prefixes on which agreement was already higher. Fixed-reference loss, fresh-policy loss, and downstream capability are therefore separate measurements. A finite-k loss measures the stated retained-support surrogate, not unbiased full-vocabulary KL.

Finite prompt streams. The independent micro-batch covariance model is exact for the independent diagnostic draws, conditional on the fixed state. Joint training consumes prompts without replacement, so its batch means are correlated. To state the correction, consider a uniform sample of $n_i = bm_i$ distinct prompts from a remaining task pool of size N_i. At a fixed student state, let $u_i(x)$ be the response-averaged expected gradient for prompt x, $R_i(x)$ the conditional rollout-gradient covariance, $S_{u,i}$ the finite-population covariance of u_i with denominator $N_i - 1$, and $\bar{R}_i$ the population average of $R_i(x)$. With independent response randomness conditional on prompts,

$$\operatorname{Cov}(\bar{g}_i) = \frac{1}{n_i}\left(1 - \frac{n_i}{N_i}\right) S_{u,i} + \frac{1}{n_i} \bar{R}_i. \tag{23}$$

The first term follows from finite-population sampling; the second follows by conditioning on the selected prompts and averaging their independent rollout noise. For independent task pools, the covariance of A is the sum of these matrices weighted by w_i^2. The local progress and Cantelli arguments continue to hold with this covariance, but the simple Σ_i/m_i formula is then an approximation. GPAS uses that approximation during training. Logs must record remaining pool sizes so its accuracy can be checked; the finite-population correction is small only when the sampled fraction and the relevant prompt-mean contribution make it small. Exhausting and reshuffling a specialist's stream also changes the conditional prompt pool and must be recorded.

A.5 Ideal sequence-level teachers and the scope of RL equivalence

For this subsection only, consider complete response distributions. Task i has reward $R_i(x, y)$, temperature $\beta_i > 0$, and the shared reference $\pi_0(y \mid x)$. Assume the required expectations and derivatives exist, the policies have compatible support, and the partition function below is finite. Define

$$\begin{aligned} J_i^{\mathrm{RL}}(\theta) &= \mathbb{E}_{x\sim\mathcal{D}_i, y\sim\pi_\theta} R_i(x, y) - \beta_i \mathbb{E}_x \, \mathrm{KL}(\pi_\theta(\cdot \mid x) \| \pi_0(\cdot \mid x)), \\ T_i^\star(y \mid x) &= \frac{\pi_0(y \mid x) \exp(R_i(x, y)/\beta_i)}{Z_i(x)}, \qquad Z_i(x) = \mathbb{E}_{y\sim\pi_0} \exp(R_i(x, y)/\beta_i). \end{aligned} \tag{24}$$

The ideal teacher is the unrestricted KL-regularized RL optimum. Substitution of $\log T_i^\star = \log \pi_0 + R_i/\beta_i - \log Z_i$ gives

$$\begin{aligned} \mathcal{F}_i^\star(\theta) &:= -\mathbb{E}_x \, \mathrm{KL}(\pi_\theta \| T_i^\star) = \frac{J_i^{\mathrm{RL}}(\theta)}{\beta_i} - \mathbb{E}_x \log Z_i(x), \\ \nabla \mathcal{F}_i^\star &= \frac{\nabla J_i^{\mathrm{RL}}}{\beta_i}. \end{aligned} \tag{25}$$

Sharing an initialization does not establish these assumptions. In particular, an approximately trained teacher need not be the Boltzmann distribution for its stated reward and temperature.

For normalized target RL weights w_i, normalized distillation weights

$$\alpha_i = \frac{w_i \beta_i}{\sum_j w_j \beta_j} \tag{26}$$

make $\sum_i \alpha_i \nabla \mathcal{F}_i^\star$ a positive global multiple of $\sum_i w_i \nabla J_i^{\mathrm{RL}}$. The coefficient condition $\alpha_i/\beta_i = c w_i$ is also necessary if this identity must hold for arbitrary task fields. It need not be necessary at a particular checkpoint where those fields are linearly dependent. For teachers queried on the same prompt, their unrestricted reverse-KL mixture has optimum $T_\alpha(y \mid x) \propto \prod_i T_i^\star(y \mid x)^{\alpha_i}$: expansion gives $\sum_i \alpha_i \, \mathrm{KL}(\pi \| T_i^\star) = \mathrm{KL}(\pi \| T_\alpha) - \log Z_\alpha(x)$, where $Z_\alpha(x) = \sum_y \prod_i T_i^\star(y \mid x)^{\alpha_i}$. Thus changing objective weights can change the target distribution.

These sequence-level identities do not identify the implemented frozen-prefix update with an RL gradient. Differentiating a sequence KL includes derivatives of the student's prefix visitation law. Equation 3 freezes that law, averages positions within responses, and uses a finite retained support. Each distinction must be accounted for before transferring an objective equivalence to an update rule.

Positive task-wise scaling preserves gradient cosines and scales raw inner products by $1/(\beta_i \beta_j)$. If every task field tends to zero, Cauchy–Schwarz implies that all cross-task inner products tend to zero. The converse is false: $(1, 0)^\top$ and $(0, 1)^\top$ have zero inner product without either field vanishing. Aggregate stationarity is also insufficient: gradients 1 and -1 cancel under equal weights but have inner product -1. Finally, opposing fields can shrink while their cosine remains -1. Raw inner-product contraction alone therefore establishes neither improved angular alignment nor task-wise convergence.

A.6 Centered signals for the implemented retained-support loss

Fix $\mathcal{R}_i^t$. Its response averaging induces a prefix law ν_i^t: draw $(x, y) \sim \mathcal{R}_i^t$, choose a position uniformly within the nonempty response, and retain its prefix c. Let $s_\theta(c, a) = \nabla_\theta \log p_\theta(a \mid c)$ and

$$z_i(c, a) = \mathbf{1}_{\{a \in S_i(c)\}} \log \frac{q_i(a \mid c)}{p_\theta(a \mid c)}. \tag{27}$$

All probabilities retain full-vocabulary normalization. Since $\nabla[p \log(p/q) - p + q] = \log(p/q) \nabla p$ for fixed q, the descent field of Equation 3 is

$$h_i^{(k)}(\theta) := -\nabla L_i^t(\theta) = \mathbb{E}_{c\sim\nu_i^t, a\sim p_\theta}[z_i(c, a) s_\theta(c, a)]. \tag{28}$$

In particular, $h_i^{(k)}(\theta_t) = -\mu_i$. The score identity $\mathbb{E}_{a\sim p_\theta}[s_\theta(c,a) \mid c] = 0$ implies

$$\begin{aligned} h_i^{(k)}(\theta) &= \mathbb{E}[(z_i - \mathbb{E}[z_i \mid c])s_\theta], \\ \|h_i^{(k)}(\theta)\| &\le \sqrt{\mathbb{E}_c \operatorname{Var}_{a\sim p_\theta}(z_i \mid c)}\sqrt{\mathbb{E}\|s_\theta\|^2}. \end{aligned} \tag{29}$$

The inequality follows from Cauchy–Schwarz, assuming finite second moments. It bounds the update from above; a large centered signal does not ensure a large update, because its correlation with the score also matters.

The centering applies to the entire masked signal z_i. It does not permit arbitrary baseline subtraction inside the retained support, since

$$\mathbb{E}_{a\sim p_\theta}[\mathbf{1}_{\{a\in S_i(c)\}} s_\theta(c,a) \mid c] = \nabla_\theta p_\theta(S_i(c) \mid c), \tag{30}$$

which is generally nonzero. In particular, prefix-only partition terms in a reward-to-teacher log-ratio decomposition cannot simply be discarded from the retained-support field.

Teacher mismatch can be isolated when the prefix law and retained support S are held fixed. For positive teacher distributions q and $\widetilde{q}$, define $r(c,a) = \mathbf{1}_{\{a\in S(c)\}} \log(q(a \mid c)/\widetilde{q}(a \mid c))$. Their descent fields satisfy

$$\begin{aligned} h(q;S) - h(\widetilde{q};S) &= \mathbb{E}[(r - \mathbb{E}[r \mid c])s_\theta], \\ \|h(q;S) - h(\widetilde{q};S)\| &\le \sqrt{\mathbb{E}_c \operatorname{Var}(r \mid c)}\sqrt{\mathbb{E}\|s_\theta\|^2}. \end{aligned} \tag{31}$$

Thus the field error depends on teacher differences on the student's evaluated states and their correlation with its score. A standalone teacher capability score does not determine this error. If the two teachers instead select different supports S and $\widetilde{S}$, the difference additionally contains $\mathbb{E}[(\mathbf{1}_S - \mathbf{1}_{\widetilde{S}}) \log(\widetilde{q}/p_\theta)s_\theta]$; the common-support residual alone does not describe that comparison.

A.7 Local curvature and slow progress without interference

Keep prefix laws and teacher supports fixed, and suppose a common parameter $\theta^\dagger$ matches each teacher on every retained token almost surely under its prefix law. At this point, each retained-support loss has zero gradient. Differentiating once more and using $\log(p_{\theta^\dagger}/q_i) = 0$ on the support gives

$$H_i^\dagger := \nabla^2 L_i^t(\theta^\dagger) = \mathbb{E}_{c\sim\nu_i^t} \sum_{a\in S_i(c)} p_{\theta^\dagger}(a \mid c) s_{\theta^\dagger}(c,a) s_{\theta^\dagger}(c,a)^\top \succeq 0. \tag{32}$$

For full vocabulary this is the conditional Fisher matrix averaged over prefixes. For finite k it is a partial score second moment; matching the retained probabilities need not determine the distribution within the omitted tail. Away from a matching point, the Hessian need not equal this matrix or be positive semidefinite.

For $e = \theta - \theta^\dagger$, a local quadratic approximation with fixed weights and fixed D gives

$$\dot{e} = -DH_w^\dagger e, \qquad H_w^\dagger = \sum_i w_i H_i^\dagger, \qquad e(t) = \exp(-DH_w^\dagger t)e(0). \tag{33}$$

This solves the linearized flow, not the general nonlinear training trajectory. For positive diagonal D, the decay rates are determined by $D^{1/2}H_w^\dagger D^{1/2}$; null directions need not contract. The constant-matrix solution does not generally apply when weights, prefix laws, or adaptive optimizer state change.

For an exact two-coordinate example, take $L_1(x,y) = x^2/2$ and $L_2(x,y) = 0.1y^2/2$, with $D = I$ and $w_1 = w_2 = 1/2$. The task gradients are orthogonal everywhere and there is no sampling noise. Nevertheless,

$$\begin{aligned} x(t) = x(0)e^{-t/2}, &\qquad y(t) = y(0)e^{-0.05t}, \\ \frac{L_1(t)}{L_1(0)} = e^{-t}, &\quad \frac{L_2(t)}{L_2(0)} = e^{-0.1t}, \end{aligned} \tag{34}$$

where normalized losses assume nonzero initial coordinates. The second task takes ten times as long to reach the same fractional loss reduction. More precise gradient estimates cannot improve this noiseless flow. This example establishes a limitation of interference and variance diagnostics; it does not identify the cause of slow MOPD training.

A.8 Minimum completion time in a separable exposure model

Consider a separate model in which nonnegative task coefficients α_i sum to one and directly set the mean learning rates:

$$\ell_i(t) = \ell_i(0)\exp(-2\alpha_i\gamma_i t), \qquad \gamma_i > 0. \tag{35}$$

Time here measures total task exposure with equal unit costs. A factorization $\gamma_i = \kappa_i\lambda_i$, with $\kappa_i, \lambda_i > 0$, can distinguish a phenomenological signal scale from a curvature scale, but the model depends only on their product. It does not derive either factor from MOPD training.

For positive loss thresholds δ_i, define

$$B_i = \begin{cases} 0, & \ell_i(0) \le \delta_i, \\ \log(\ell_i(0)/\delta_i)/(2\gamma_i), & \ell_i(0) > \delta_i. \end{cases} \tag{36}$$

Reaching all thresholds by T is equivalent to $\alpha_i T \ge B_i$ for every task. Summing yields $T \ge \sum_i B_i$. If this sum is positive, the static choice

$$T^\star = \sum_i B_i, \qquad \alpha_i^\star = \frac{B_i}{\sum_j B_j} \tag{37}$$

attains the bound. If all $B_i = 0$, no training is required: $T^\star = 0$ and any feasible coefficients suffice.

This result assumes independent exponential loss dynamics and unconstrained continuous coefficients. It is not a minimum-time guarantee for shared neural parameters, unequal compute costs, or AdamW. Moreover, its α_i change the mean field, whereas GPAS changes sample counts m_i while preserving the chosen weights w_i in Equation 6. The two allocation problems have different objectives; this completion-time formula does not establish optimal training time for GPAS.

A.9 Mean progress and directional reliability

All quantities below are conditional on the same checkpoint, chosen counts, and fixed positive diagonal D. Suppose the $g_{j,s}$ are independent with mean μ_j and covariance Σ_j. Then

$$\mathbb{E}A = \mu_w, \qquad \Sigma(m) := \mathrm{Cov}(A) = \sum_j \frac{w_j^2}{m_j}\Sigma_j. \tag{38}$$

Linearity gives $\mathbb{E}X_i = \mu_i^\top D\mu_w = a_i$, and the variance of a linear functional gives $\mathrm{Var}(X_i) = \mu_i^\top D\Sigma(m)D\mu_i = v_i(m)$. The first-order loss change is $-\eta X_i$. Consequently, if $a_i \le 0$, its expectation is nonnegative for every admissible count vector. Additional samples can change the variance and finite-step curvature contribution, but cannot make this expected first-order change negative at the fixed mean and D.

For completeness, write $Z_i = a_i - X_i$, so that $\mathbb{E}Z_i = 0$ and $\mathbb{E}Z_i^2 = v_i$. For any $\lambda \ge 0$ and $a_i > 0$, Markov's inequality implies

$$\Pr(X_i \le 0) = \Pr(Z_i \ge a_i) \le \frac{\mathbb{E}(Z_i+\lambda)^2}{(a_i+\lambda)^2} = \frac{v_i+\lambda^2}{(a_i+\lambda)^2}. \tag{39}$$

Choosing $\lambda = v_i/a_i$ proves $\Pr(X_i \le 0) \le v_i/(v_i + a_i^2)$, including the zero-variance case. The union bound gives Equation 13 without requiring the X_i to be independent. These random variables usually share the same A and are dependent.

The matrix K is a Gram matrix in the inner product induced by D; it is positive semidefinite, but Kw can have nonpositive entries. The signs of those entries determine expected task progress for the chosen mixture. Neither an off-diagonal negative entry nor a local nonpositive margin establishes that the capability targets are globally unattainable. The analysis also permits successful trajectories with temporary task regressions.

A.10 A sufficient local sampling budget

Assume $a_i > 0$ for every domain. For positive fractions q_j summing to one, suppose the counts $m_j = Gq_j$ are integers and satisfy any applicable bounds. Define

$$B_i(q) = \sum_j \frac{w_j^2\mu_i^\top D\Sigma_j D\mu_i}{q_j a_i^2}. \tag{40}$$

Then $v_i(m)/a_i^2 = B_i(q)/G$. For any $0 < \delta < 1$, the sufficient condition

$$G \geq \frac{M - \delta}{\delta} \max_i B_i(q) \tag{41}$$

ensures $\Pr(\exists i : X_i \leq 0) \leq \delta$. Indeed, each Cantelli term is $B_i/(G + B_i)$, which is at most δ/M under the stated condition; summing proves the claim. If all $B_i = 0$, the sampled first-order progress is deterministic.

This is a sufficient condition on known population quantities. Rounding counts requires recomputing the bound for the resulting allocation. Count caps may prevent the budget from being attained. The condition also omits estimation uncertainty, finite-population corrections, curvature, and state evolution; it is not an operational guarantee for AdamW or a requirement for long-run capability integration.

A.11 Trace variance and exact integer allocation

By Equation 38,

$$V(m) = \operatorname{tr}(D\Sigma(m)D) = \sum_i \frac{w_i^2 \operatorname{tr}(D\Sigma_i D)}{m_i} = \sum_i \frac{w_i^2 e_i}{m_i}. \tag{42}$$

For the positive semidefinite matrix $Q = D\Sigma(m)D$, $\mu_i^\top Q \mu_i \leq \|\mu_i\|_2^2 \lambda_{\max}(Q) \leq \|\mu_i\|_2^2 \operatorname{tr}(Q)$, proving Equation 15. Equation 17 is the sum of coordinate-wise sample variances and is unbiased for e_i under independent sampling. The moving average is an allocation estimate, not an unbiased estimate of a changing current-state covariance.

Write $c_i = w_i\sqrt{e_i}$. Cauchy–Schwarz gives

$$\left(\sum_i m_i\right)\left(\sum_i \frac{c_i^2}{m_i}\right) \geq \left(\sum_i c_i\right)^2. \tag{43}$$

When all $c_i > 0$, equality holds at $m_i = Gc_i/(\sum_j c_j)$, yielding the continuous per-step rule without count bounds. Positive lower and upper bounds lead to the corresponding clipped continuous rule by the Karush–Kuhn–Tucker conditions. Zero-noise components are handled by the bounds and remaining budget. We do not claim that rounding this continuous solution solves the integer problem. Both GPAS modes evaluate their estimated objectives on every $m \in \mathcal{M}$ as defined in Appendix A.2.

With positive marginal times and zero fixed step time,

$$\left(\sum_i m_i \tau_i\right)\left(\sum_i \frac{c_i^2}{m_i}\right) \geq \left(\sum_i c_i\sqrt{\tau_i}\right)^2, \tag{44}$$

with equality at $m_i \propto c_i/\sqrt{\tau_i}$ when this continuous allocation is feasible. Positive fixed time and integer bounds are handled by enumeration. As C grows on the finite feasible set, minimizing $CV(m) + (\sum_i m_i\tau_i)V(m)$ eventually selects a minimum-variance allocation, with time breaking any remaining variance ties. The product criterion alone does not establish a learning-rate or time-to-capability improvement.

For the study's Uniform counts of four and cap of eight, $V(m) \geq \sum_i w_i^2 e_i/8 = V_{\text{Uniform}}/2$. This proves the stated factor-of-two ceiling under the independent, fixed-state model for any feasible allocation. All these statements concern the same means and covariances, not different training states reached by different schedules.

A.12 Trace-optimal allocation need not preserve task directions best

Consider two tasks with $D = I$, $w_1 = w_2 = 1/2$, and $\mu_1 = \mu_2 = (1, 0)^\top$. Let

$$\Sigma_1 = \begin{pmatrix} 1 & 0 \\ 0 & 80 \end{pmatrix}, \qquad \Sigma_2 = \begin{pmatrix} 9 & 0 \\ 0 & 0 \end{pmatrix}. \tag{45}$$

Both progress margins equal one. For $G = 40$ with positive counts and no additional caps, the trace noises $(e_1, e_2) = (81, 9)$ give the GPAS optimum $(m_1, m_2) = (30, 10)$ and

$$v_1 = v_2 = \frac{1}{4}\left(\frac{1}{30} + \frac{9}{10}\right) = \frac{7}{30}. \tag{46}$$

The allocation $(10, 30)$ instead gives $v_1 = v_2 = \frac{1}{4}(1/10 + 9/30) = 1/10$. The latter allocation minimizes the common directional variance: its relevant standard deviations are in the ratio $1 : 3$. The trace variances are respectively $9/10$ and $21/10$, so the trace and directional objectives prefer different allocations.

If the independent micro-batch noise is Gaussian, then $X_1 = X_2$ is Gaussian with mean one and the stated variance. Its exact non-descent probability is $\Phi(-1/\sqrt{v_1})$, where Φ is the standard normal distribution function. Thus $(10, 30)$ also has lower exact failure probability, not merely a lower bound. The discrepancy comes from task 1's second-coordinate variance, which is orthogonal to both task means. This is an analytical counterexample; its two-task budget is separate from the four-domain training protocol.

A.13 Finite-step progress and the optimizer approximation

Suppose L_i^t is $\mathcal{L}_i$-smooth along every update segment under consideration. For the idealized update,

$$L_i^t(\theta_t - \eta DA) - L_i^t(\theta_t) \le -\eta X_i + \frac{\eta^2 \mathcal{L}_i}{2}\|DA\|_2^2. \tag{47}$$

Taking expectations gives

$$\mathbb{E}[\Delta L_i^t] \le -\eta a_i + \frac{\eta^2 \mathcal{L}_i}{2}\left(\|D\mu_w\|_2^2 + V(m)\right). \tag{48}$$

Consequently, when all margins are positive, a sufficient condition for every expected task-loss change to be negative is

$$0 < \eta < \min_i \frac{2a_i}{\mathcal{L}_i(\|D\mu_w\|_2^2 + V(m))}, \tag{49}$$

with an infinite threshold for a zero smoothness constant. A realized first-order descent sign by itself is insufficient: Equation 47 additionally requires $X_i > \eta \mathcal{L}_i \|DA\|_2^2/2$ to certify a strict decrease.

A probability version requires control of update magnitude as well. If $\Pr(\|DA\|_2 > R) \le \varepsilon_R$ and $d_i = a_i - \eta \mathcal{L}_i R^2/2 > 0$ for all i, Cantelli and a union bound give

$$\Pr(\exists i : \Delta L_i^t \ge 0) \le \min\left\{1, \varepsilon_R + \sum_i \frac{v_i(m)}{v_i(m) + d_i^2}\right\}. \tag{50}$$

The event of a nondecrease with $\|DA\|_2 \le R$ implies $X_i \le \eta \mathcal{L}_i R^2/2$ for some i, which proves the bound. These assumptions need to be verified before interpreting the expression as a finite-step guarantee.

A second-order expansion states how covariance direction and curvature enter more precisely. Let $H_i = \nabla^2 L_i^t(\theta_t)$. If the Hessian is κ_i-Lipschitz on all update segments and $\mathbb{E}\|DA\|_2^3 < \infty$, then

$$\begin{aligned} \mathbb{E}[\Delta L_i^t] &= -\eta a_i + \frac{\eta^2}{2}\left[(D\mu_w)^\top H_i (D\mu_w) + \operatorname{tr}(H_i D \Sigma(m) D)\right] + \mathcal{E}_i, \\ |\mathcal{E}_i| &\le \frac{\kappa_i \eta^3}{6}\mathbb{E}\|DA\|_2^3. \end{aligned} \tag{51}$$

This follows by Taylor's theorem and the quadratic-form bias–variance identity. The curvature-weighted covariance term can have either sign when the Hessian is indefinite. The smoothness bound replaces this directional dependence with the conservative trace term.

For the weighted frozen objective F_t, the same argument gives, with $\mathcal{L}$ a smoothness constant,

$$\mathbb{E}F_t(\theta_t - \eta DA) \le F_t(\theta_t) - \eta \mu_w^\top D \mu_w + \frac{\eta^2 \mathcal{L}}{2}\left(\|D\mu_w\|_2^2 + V(m)\right). \tag{52}$$

Improving this weighted bound does not ensure progress for every task. The idealized update is also not the actual AdamW update: momentum, current-gradient second moments, clipping, and decay can change its mean and direction. Allocation can alter these states over training. The common-checkpoint experiment therefore compares frozen-D predictions with actual optimizer branches using identical saved states.

A.14 CONDITIONAL EXTENSION: ALLOCATION FOR DIRECTIONAL UNCERTAINTY

If measurements show that trace noise is a poor proxy for relevant uncertainty, a possible extension is to choose counts by

$$\min_{m \in \mathcal{M}} \max_{1 \le i \le M} \frac{v_i(m)}{a_i^2}, \qquad a_i > 0 \quad \text{for every } i. \tag{53}$$

With known positive margins and covariances, each normalized variance is a sum of nonnegative multiples of $1/m_j$. It is convex on positive counts; their maximum is convex, and the sum and box constraints are convex in the continuous relaxation. With four tasks the integer problem can again be solved by finite enumeration. This criterion minimizes the worst normalized directional variance, and hence the worst Cantelli bound, under the stated local model. It need not minimize the true joint failure probability or long-run capability loss.

If any margin is nonpositive, the criterion does not resolve joint progress. Excluding that domain would conceal the unresolved task tradeoff. Changing weights, the update direction, or parameter sharing requires a separate intervention. Estimating directional covariances also adds computation: independent calibration and trial samples, or historical estimates fixed before the trial, are needed to test the predictions without reusing the same noise optimistically. The proposed study first tests GPAS and its trace statistic. This extension is a research hypothesis, not an implemented or validated algorithm.

A.15 TIMING AND PLANNED TRAINING RECORDS

Task blocks record rollout, teacher-scoring, backward, and statistics time, together with response lengths and peak memory. The marginal per-micro-batch estimates form τ_i; synchronization, optimizer work, and other count-independent step operations form C. Their EMA estimates use decay 0.9. We compare predicted and observed step times and evaluate efficiency with actual elapsed time on both reserved GPUs. Evaluation compute is reported separately. If a deployment overlaps task blocks, its timing predictor should represent that execution schedule.

Checkpoints retain optimizer state, task stream positions, counts, and noise and time averages. The planned traces include per-task held-out loss, prompt consumption, response length and truncation, micro-batch counts, noise estimates, and measured step time. **[TODO: Add the artifact location and populate the configuration and empirical figures from these records.]**